Bursty infectiousness as a hidden driver of epidemic variability

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Abstract

Epidemic trajectories are profoundly shaped by random, individual-level effects. Superspreading can make epidemics less likely to establish, but more explosive when they do; and chance events during the cryptic early stages of an outbreak can durably shift its overall timing.[1, 2] Accounting for such effects is vital for public health preparedness: while a good understanding of an epidemic's expected trajectory can be achieved using deterministic, population-level models, the variation around that mean — including best- and worst-case scenarios — hinges upon individual variation in transmission capacity and timing.

One such factor that has so far received little attention is the concentration of individual infectiousness, or its "burstiness". Evidence from SARS-CoV-2 and influenza suggests that the window of meaningful infectiousness for those viruses may be short — even less than a day — despite the fact that the infectiousness window is typically quoted as lasting 6-8 days for influenza and 9-12 days for SARS-CoV-2.[3, 4, 5] These longer spans reflect the generation interval distribution, $g(\tau)$, a population-level average of the times τ between primary and secondary infections. Its individual-level counterpart, the infectiousness kernel[6, 7], describes the distribution of transmission times for a single infector, and need not resemble $g(\tau)$: $g(\tau)$ can be broad simply because different individuals' infectiousness windows fall at different times, even if each individual is only briefly infectious. If the infectiousness kernel is in reality much more concentrated than $g(\tau)$, this could impact everything from how individuals gauge their own risk of transmitting to how epidemics unfold overall.

Although individual-based simulations can in principle accommodate heterogeneous infectiousness timing, standard analytical frameworks operate on the population-level $g(\tau)$ directly and thus are blind to burstiness by construction. More fundamentally, no existing framework has been designed to isolate burstiness as a free parameter while holding $g(\tau)$ and R_0 fixed, making it impossible to attribute observed epidemic dynamics specifically to burstiness. Empirical evidence is likewise limited: resolving the infectiousness kernel requires densely sampled, individual-level transmission data that is rarely available, and what data exist are subject to biases that can both inflate and obscure the apparent degree of burstiness. Whether burstiness has epidemiologically meaningful consequences therefore remains an open question.

Here, we address that gap by systematically examining the impact of bursty infectiousness on epidemic dynamics. We introduce the burstiness parameter, $\psi \in [0, 1]$, which governs the width of the individual infectiousness kernel — specifically, the fraction of $g(\tau)$'s variance attributable to within-infector variation in transmission timing (**Supplementary Methods**). At the $\psi = 0$ extreme (maximum burstiness), each infector transmits at a single moment, with that moment drawn from $g(\tau)$. At the $\psi = 1$ extreme, each infector's secondary infections are distributed according to $g(\tau)$ itself. Critically, varying ψ has no impact on the fundamental population-level epidemiological quantities, $g(\tau)$ and R_0. In this respect, ψ stands in the same relation to $g(\tau)$ as the overdispersion parameter k stands to R_0 [1]: a measure of individual-level variation that is invisible to population-level averages yet shapes epidemic dynamics. We show that burstiness drives an underappreciated form of superspreading, inflates variability in outbreak timing, shapes how isolation and gathering-size restrictions affect epidemic dynamics, and complicates estimation of key epidemiological parameters.

To embed ψ within a framework suitable for analyzing its epidemiological properties, we introduce the Gamma burst model. The model assumes $g(\tau)$ is (approximately) Gamma-distributed with shape α and rate β. At the individual level, the infectiousness kernel is a Gamma$(\psi\alpha, \beta)$ distribution — the "burst" — following a latent period of length $l \sim$ Gamma$((1-\psi)\alpha, \beta)$ (**Figure XX**). Since both components are Gamma-distributed with the same rate, their sum recovers Gamma$(\alpha, \beta) = g(\tau)$, ensuring that any value of ψ yields the same population-level generation interval distribution. The Gamma burst model has biological grounding in within-host viral dynamics: stochastic viral replication early in infection creates between-individual variation in the onset of detectable viral growth (the variable latent period), while subsequent shedding kinetics are more consistent across individuals (the infectiousness burst)[8].

Empirical estimates of ψ. To gauge the potential range of burstiness across pathogens, we estimated ψ using published transmission trees with serial intervals for COVID-19, MERS, measles, Ebola, smallpox, pneumonic plague, norovirus, and hepatitis A [9] (**Methods**). The resulting estimates spanned a wide range (**Figure XX**): measles and pneumonic plague showed the strongest evidence of burstiness (95% credible interval for ψ: [0.00, 0.11] and [0.00, 0.20], respectively), while smallpox and MERS showed evidence of more diffuse infectiousness (95% credible interval for ψ: [xx,xx] and [xx,xx], respectively).

These estimates should be regarded with caution. Multiple biases can distort estimates of ψ in either direction: uncertainty in infection timing can artificially disperse observed infection times, biasing ψ upward; whereas an emphasis on recording superspreading events while ignoring less explosive transmission could bias ψ downward. Moreover, since $g(\tau)$'s variance must reflect at least some between-individual variation in infectiousness timing, values of ψ near 1 are likely overestimates. Nevertheless, the range of estimates — along with independent viral kinetics evidence suggesting narrow infectiousness windows for influenza and SARS-CoV-2 [3] — indicates that ψ varies across pathogens, and that bursty infectiousness reflects a genuine epidemiological feature rather than a measurement artifact.

For the remainder of our analyses, we illustrate our main findings using three benchmark pathogens spanning a range of reproduction numbers and generation interval shapes (**Figure XX**): influenza ($R_0 = 2$), SARS-CoV-2 omicron ($R_0 = 6$), and measles ($R_0 = 12$). For each, we examine the impact of burstiness across its full range ($\psi \in [0, 1]$), though the methods and findings are general.

Bursty infectiousness drives coincidence superspreading. Superspreading happens when a person infects significantly more than the expected number of secondary cases (R_0). Typically, superspreading is assumed to arise from pesistent individual-level traits: either a person has more than the average number of contacts, or sheds more than the typical amount of pathogen. The mechanism matters: contact-driven and infectiousness-driven superspreading produce qualitatively different epidemic dynamics.[10]

A third type of superspreading can arise when, by chance, an infectiousness burst happens to align with a brief period of high contacts. This "coincidence superspreading" is fundamentally different from other types: it does not arise from any persistent individual trait, and therefore cannot be targeted by individual-level interventions. It thus represents a floor on overdispersion that persists even after all "super-shedder" and "super-contactor" heterogeneity has been accounted for.

In general, burstier infectiousness leads to more coincidence superspreading. This can be illustrated by considering a simple periodic model for contact rates: $c(t) = 1 - \rho\cos(2\pi t/\omega)$, where $\rho \in [0, 1]$ is the amplitude and $2\pi/\omega$ is the period of the contact variation, relative to a baseline contact rate of $\bar{c} = 1$ (**Supplementary Methods**). This $c(t)$ captures variation over time, but not across individuals: each person is assumed to be identical in every way except for the timing of their infection. Under the Gamma burst model, and assuming that infections occur at uniform-random times relative to the contact process, we can calculate the overdispersion in secondary infection counts in terms of the Lloyd-Smith parameter, k:

$$\mathrm{OD} \equiv 1/k = \frac{\rho^2}{2}\left(1 + \left(\frac{2\pi\bar{g}}{\alpha\omega}\right)^2\right)^{-\psi\alpha} \tag{1}$$

Here, α is the shape parameter of the generation interval distribution and $\bar{g} = \alpha/\beta$ is the mean generation time. Smaller values of k correspond to higher overdispersion (OD), with $k \to \infty$ denoting minimum overdispersion (OD $\to 0$, equivalence between the mean and variance of secondary infection counts).

OD is maximized by bursty infectiousness ($\psi \to 0$). The sensitivity of OD to ψ depends on $\eta = 2\pi\bar{g}/(\alpha\omega)$: when $\eta \gg 1$, which happens for example when the contacts vary quickly relative to the mean generation time ($\omega \ll \bar{g}$), virtually all infection kernels average over the entire contact process, yielding little overdispersion (OD $\to 0$) regardless of ψ. Alternatively, when $\eta \ll 1$, which can happen with contacts vary slowly relative to the mean generation time ($\omega \gg \bar{g}$), all infection kernels — whether narrow or broad — only "see" a small part of the contact variation, causing maximum overdispersion (OD $\to \rho^2/2$) that is insensitive to ψ. The impact of ψ is thus most visible when $\eta \approx 1$, in which case broad infectiousness ($\psi \to 1$) causes the infection kernel to effectively average over the contact variation, while bursty infectiousness ($\psi \to 0$) translates the contact process' variation directly into elevated overdispersion. For acute respiratory pathogens, where $\bar{g}$ may range from roughly 3 to 12 days, weekly-varying contact patterns are most relevant for ψ-related coincidence superspreading; daily variation gets averaged out, while monthly variation contributes to overdispersion regardless of ψ. Empirical contact surveys provide show that contacts vary meaningfully on these timescales: the POLYMOD and BBC Pandemic studies document XX% - XX% fewer contacts on weekends than weekdays across European populations, corresponding to $\rho \approx 0.1 - 0.2$ at $\omega = 7$ days.

For the three benchmark pathogens (influenza, SARS-CoV-2 omicron, and measles), simulations confirm these analytic predictions across a range of contact amplitudes and periods, and showed that the resulting overdispersion translates into systematically higher epidemic extinction probabilities (**Figure XX**). Similar patterns emerged for stochastic contact processes (**Supplementary Methods, Supplementary Figure XX**).

Burstiness and the epidemic latent period. Like infections, epidemics also have a "latent period" — the span between the introduction of the first case and when the outbreak becomes established. The epidemic latent period's variability depends on early-outbreak stochastic effects.[2] Bursty infectiousness inflates the variability of this latent period, leading to meaningful shifts in outbreak timing.

To illustrate this, we can describe the early exponential growth phase of an epidemic as a branching process [2]:

$$Z(t) \approx We^{rt} \tag{2}$$

where $Z(t)$ is the cumulative number of cases at time t, r is the epidemic growth rate, and W is a random initial condition capturing early-outbreak stochasticity. Large values of W correspond to random successful early transmission events, inflating the effective initial condition. With a change of variables, this random initial condition can be expressed as a random time shift ς, so that the branching process can be expressed as $Z(t) \approx e^{r(t+\varsigma)}$.[2]

Bursty infectiousness inflates $\mathrm{Var}[W]$, which makes epidemic timing more variable (increases $\mathrm{Var}[\varsigma]$) and shifts the typical epidemic slightly later (decreases $E[\varsigma]$). Formally,

$$\mathrm{Var}[W] \propto z(R_0, \alpha)^{(1-\psi)\alpha} \tag{3}$$

where z is a quantity that depends on R_0 and α; it is increasing in R_0 and is greater than 1 when $R_0 > 1$. Thus, bursty infectiousness ($\psi \to 0$) maximizes $\mathrm{Var}[W]$, with the largest effect when R_0 is high. If, following prior work, we assume W (conditional on epidemic survival) is approximately Gamma-distributed, this translates into more variable and later-shifted epidemic timing (**Supplementary Methods**).

Simulations illustrate the impact of bursty infectiousness on epidemic timing and its relationship with R_0. Survival curves depicting the time to reach 5% total prevalence show marked differences across ψ-values for SARS-CoV-2 omicron and measles (**Figure XX**). For example, the fraction of simulated measles outbreaks that reached 5% prevalence by week 5 was xx% at the bursty extreme ($\psi = 0$) *vs.* xx% at the smooth extreme ($\psi = 1$), consistent with burstiness driving more variable and late-shifted epidemic timing. For influenza, which has a lower R_0, differences in epidemic timing across ψ-values were hardly discernible (**Figure XX**).

Common interventions are sensitive to bursty infectiousness. Detect-and-isolate (D&I) interventions aim to identify and isolate individuals who are, or are about to be, infectious, and thus to limit onward transmission. Their efficacy depends on the timing of detection relative to infectiousness.[11] When detectability is linked to infectiousness — as is realistic for symptom-based or diagnostic testing-based detection — bursty infectiousness can cause the effectiveness of D&I interventions to change dramatically with small shifts in detection time.

To see this, we calculated the testing effectiveness (TE), defined as the expected proportion of transmission that a D&I intervention prevents.[12] We calculated TE under two detection mechanisms: symptom-based detection, where symptoms arise at a Normal(δ, σ)-distributed offset from peak infectiousness; and testing-based detection, where tests are administered at fixed intervals of length Δ days, and the first test falling within $-\delta_{pre}$ days before through δ_{post} days after peak infectiousness triggers isolation (**Methods**). For both detection mechanisms, TE declined as the typical detection time shifted later, and the sharpest drops corresponded to the narrowest infectiousness kernels (**Figure XX**). This led to substantial ψ-related differences in TE: for example, for symptom-based screening against SARS-CoV-2 omicron with $\delta = xx$ days and $\sigma = xx$, TE ranged from ≈ 0 for $\psi = 0$ to xx for $\psi = 1$.

Beyond TE, D&I interventions produce two additional ψ-dependent effects. First, D&I interventions truncate the effective generation interval distribution, since early transmissions are more likely to succeed than late ones. This truncation is maximized for broad infectiousness kernels, where detection reliably removes late transmission; and it disappears as $\psi \to 0$, since detection is all-or-nothing and surviving transmission attempts succeed as if no intervention were in place. Second, D&I interventions can create overdispersion in secondary case counts: stochastic variation in detection time can cause outsize differences in realized infectiousness across individuals. In the bursty extreme, a fraction $1 - \text{TE}$ of individuals retain their full infectiousness, while the rest have an infectiousness of 0 (assuming perfect isolation). These features impact the dynamics of D&I-controlled epidemics, verified by simulations: for a given TE, bursty infectiousness causes epidemics to be less likely to establish (due to the elevated overdispersion) and to grow more slowly (due to the lack of generation interval truncation) than when infectiousness is broad.

Gathering size restrictions also interact with ψ. Truncating the contact distribution at a maximum gathering size $c_{\max}$ reduces the effective reproduction number, R_{eff}, by a factor that is independent of ψ (**Supplementary Methods**). However, the truncation also reduces overdispersion, and the absolute reduction is greatest when the baseline overdispersion is highest — *i.e.*, when ψ is small — because truncating the contact distribution removes the high-contact tail that bursty infectiousness exploit. Simulations confirm that gathering-size-restricted epidemics have systematically higher extinction probabilities than R_{eff}-matched epidiemics without contact-process adjustment, with the largest discrepancies for bursty infectiousness profiles (**Figure XX**). Evaluating interventions purely in terms of their effect on R_{eff}, without accounting for ψ, can therefore be misleading.

Burstiness complicates estimation of key epidemiological parameters. In an emerging epidemic, estimating the epidemic growth rate r and the generation interval distribution $g(\tau)$ are critical tasks, both in their own right and as ingredients for calculating R_0. Bursty infectiousness makes both tasks more challenging. First, bursty infectiousness infuses excess variation into daily case counts (**Supplementary Methods; Supplementary Figure XX**), which makes estimates of r less certain, especially when R_0 is high. Using the standard Poisson generalized linear model approach to estimate r from simulated data, we found that the standard error in $\hat{r}_{\text{MLE}}$ was between xx (for influenza) and xx (for measles) times higher for $\psi = 0$ than for $\psi = 1$ (**Figure XX**).[x]

Second, bursty infectiousness causes tight correlation among secondary infection times from a single index case. This causes the effective number of draws from $g(\tau)$ produced by an index case to collapse from R_0 when $\psi = 1$ (each secondary infection time is an independent draw from $g(\tau)$) to 1 when $\psi = 0$ (all secondary infections occur simultaneously, so together they represent a single draw from $g(\tau)$). Treating serial intervals as independent, as is common practice, can produce over-confident estimates of $g(\tau)$'s parameters (**Figure XX; Supplementary Methods**).

Discussion. The burstiness of infectiousness, as captured by the parameter ψ, plays a key role in how epidemics unfold. Just as the overdispersion parameter k illustrates how not all pathogens with the same R_0 behave alike, ψ reveals that not all pathogens with the same $g(\tau)$ behave alike. Burstiness impacts overdispersion, outbreak timing, intervention effectiveness, and epidemiological parameter estimation — a wide enough range of consequences to justify treating it as a fundamental epidemiological quantity.

Our findings have various practical implications. When designing D&I interventions, burstiness can potentially be leveraged to gain significant reductions in transmission with relatively little effort: for example, a 1-day increase in testing frequency was in some cases associated with as much as a xx% predicted increase in the intervention's effectiveness. Furthermore, burstiness can impact how interventions shape population-level transmission: accounting for an intervention's reduction in R_{eff} alone can mask important, burstiness-related shifts in epidemic growth rate and extinction probability. Since burstiness can inflate variability in daily case counts and create timing correlations among secondary infections, care must be taken when estimating and interpreting epidemiological parameter values like r and $g(\tau)$.

The ingredients for our findings have been present in the literature for over a decade, but were never assembled because standard modeling frameworks integrate over individual-level timing by construction. The individual infectiousness kernel was defined by Svensson and extended by Champredon and Dushoff. Shifts in the punctuation of the generation interval distribution have demonstrated the key impact of that population-level quantity; and replacing the stepwise infectiousness profile commonly assumed by SIR and SEIR-type compartmental models with time-varying viral kinetics-informed infectiousness kernels can generate meaningful impacts on epidemic dynamics. The course of individual infectivity has been used as key input in models of contact tracing effectiveness and testing-based interventions.

A key feature of bursty infectiousness is that it introduces an often-overlooked and unintervenable mechanism of superspreading. Coincidence superspreading produces superspreading events without superspreader individuals. This formalizes the long-observed but previously *ad hoc* distinction between superspreading events and superspreaders. As a result of coincidence superspreading, some fraction of overdispersion may be structurally untargetable, setting a floor on the overdispersion k regardless of how well we identify and manage high-risk individuals.

Our findings are subject to several limitations: the Gamma burst model, while analytically tractable, may not capture the full biological variability in burstiness. The contact models we considered are stylized, not empirically driven, and they do not account for spatial or network heterogeneity. When inferring ψ, we used serial intervals as proxies for generation intervals, inflating uncertainty in our estimates.

Looking forward, a key challenge facing future work on burstiness is measurement: estimating ψ is challenging without detailed, dense contact tracing data. Some opportunities exist for incorporating other data streams, like cross-referencing with viral kinetics; but we need a better understanding of how viral load relates to infectiousness.[3] Joint inference of k and ψ remains an open question, given that they can interact with one another. More complex models — like mixture models with multiple different values of ψ belonging to the same person — could enhance the biological fidelity of the Gamma burst model, though at the cost of model complexity.

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Figure Legends

Figure 1 | **The individual infectiousness kernel and the burstiness parameter ψ.**

Figure 2 | **Empirical estimates of ψ.**

Figure 3 | **Coincidence overdispersion.**

Figure 4 | **The epidemic latent period.**

Figure 5 | **Interventions.**